Decent Coaching Classes

IX CBSE

Mathematics Worksheet | NCERT (Ganita Manjari) Class 9

Instructions: *Attempt all 15 questions. Show all steps of your working. Answers and solutions are given on the last page.*

Chapter 1: Orienting Yourself: The Use of Coordinates

1. The points A(2, 3), B(6, 3), C(6, 7) and D(2, 7) are plotted on a graph paper and joined in order.
(a) Name the figure ABCD and find its perimeter and area.
(b) Write the coordinates of the point where its diagonals meet.

Chapter 2: Introduction to Linear Polynomials

2. A taxi charges a fixed amount of ₹40 plus ₹12 for every kilometre travelled.
(a) Write the linear polynomial for the fare for x km.
(b) Find the fare for 7 km.
(c) For how many kilometres was the journey if the fare was ₹148?

3. For the linear polynomial $p(x) = 2x - 4$, find the zero of $p(x)$. Hence write the points where the graph of $y = 2x - 4$ cuts the x-axis and the y-axis.

Chapter 3: The World of Numbers

4. Express the repeating decimal $0.363636\ldots$ (the block 36 repeats forever) in the form p/q in lowest terms.

5. Simplify: $(27)^{2/3} \times (16)^{-3/4}$.

Chapter 4: Exploring Algebraic Identities

6. Using suitable identities, evaluate:
(a) 103×97
(b) $(101)^3$

7. If $x + 1/x = 5$, find the values of (a) $x^2 + 1/x^2$ and (b) $x^3 + 1/x^3$.

Chapter 5: I'm Up and Down, and Round and Round

8. A chord of a circle is 16 cm long and is at a distance of 6 cm from the centre. Find the radius of the circle.

9. An arc PQ of a circle with centre O subtends an angle of 80° at the centre. Find (a) the angle subtended by the arc at a point R on the major arc, and (b) $\angle OPQ$.

Chapter 6: Measuring Space: Perimeter and Area

10. The sides of a triangle are 13 cm, 14 cm and 15 cm. Find its area using Heron's formula and hence find the height on the side of 14 cm.

11. A rectangular park is 40 m long and 30 m wide. A path 2 m wide runs inside the park along its entire boundary. Find the area of the path.

Chapter 7: The Mathematics of Maybe: Introduction to Probability

12. A bag contains 5 red, 3 blue and 2 green balls. One ball is drawn at random. Find the probability that it is (a) red, (b) not green.

13. Two dice are thrown together. Find the probability that the sum of the numbers on the top faces is (a) 8, (b) 13.

Chapter 8: Predicting What Comes Next: Exploring Sequences and Progressions

14. For the arithmetic progression 7, 11, 15, 19, …, find the 20th term and the sum of the first 20 terms.

15. Consider the sequence 2, 6, 18, 54, … Find the common ratio, the next two terms and the 8th term.

Q1. (a) AB = 4, BC = 4 units, all sides equal and angles 90°, so ABCD is a square. Perimeter = 4 × 4 = 16 units; Area = 4 × 4 = 16 sq. units.
(b) The diagonals meet at the centre, midway between the corners: x = (2+6)/2 = 4, y = (3+7)/2 = 5. Point = (4, 5).

Q2. (a) Fare p(x) = 12x + 40.
(b) p(7) = 12 × 7 + 40 = 84 + 40 = ₹124.
(c) 12x + 40 = 148 ⇒ 12x = 108 ⇒ x = 9 km.

Q3. Zero: 2x − 4 = 0 ⇒ x = 2. Cuts the x-axis at (2, 0).
Put x = 0: y = −4, so it cuts the y-axis at (0, −4).

Q4. Let x = 0.3636… Then 100x = 36.3636… Subtract: 99x = 36 ⇒ x = 36/99 = 4/11.

Q5. $27^{2/3} = (3^3)^{2/3} = 3^2 = 9$. $16^{-3/4} = (2^4)^{-3/4} = 2^{-3} = 1/8$.
Product = 9 × 1/8 = 9/8.

Q6. (a) 103 × 97 = (100 + 3)(100 − 3) = $100^2 - 3^2$ = 10000 − 9 = 9991.
(b) $(100 + 1)^3 = 100^3 + 3(100)^2(1) + 3(100)(1)^2 + 1^3$ = 1000000 + 30000 + 300 + 1 = 1030301.

Q7. (a) $(x + 1/x)^2 = x^2 + 2 + 1/x^2 \Rightarrow 25 = x^2 + 1/x^2 + 2 \Rightarrow x^2 + 1/x^2 = 23$.
(b) $(x + 1/x)^3 = x^3 + 1/x^3 + 3(x + 1/x) \Rightarrow 125 = x^3 + 1/x^3 + 15 \Rightarrow x^3 + 1/x^3 = 110$.

Q8. The perpendicular from the centre bisects the chord, so half-chord = 8 cm and distance = 6 cm.
$r^2 = 8^2 + 6^2 = 64 + 36 = 100 \Rightarrow r = 10$ cm.

Q9. (a) The angle at the circle is half the angle at the centre: ∠PRQ = 80°/2 = 40°.
(b) OP = OQ (radii), so ∠OPQ = ∠OQP = (180° − 80°)/2 = 50°.

Q10. s = (13 + 14 + 15)/2 = 21. Area = $\sqrt{[21(21-13)(21-14)(21-15)]} = \sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7056} = 84$ cm².
Area = ½ × base × height ⇒ 84 = ½ × 14 × h ⇒ h = 12 cm.

Q11. Outer area = 40 × 30 = 1200 m². Inner rectangle = (40 − 4) × (30 − 4) = 36 × 26 = 936 m².
Area of path = 1200 − 936 = 264 m².

Q12. Total balls = 10. (a) P(red) = 5/10 = 1/2. (b) P(not green) = 1 − 2/10 = 8/10 = 4/5.

Q13. Total outcomes = 36. (a) Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) = 5 outcomes ⇒ P = 5/36.
(b) A sum of 13 is impossible ⇒ P = 0/36 = 0.

Q14. a = 7, d = 4. $a_{20} = 7 + 19 \times 4 = 83$.
$S_{20} = (20/2)(7 + 83) = 10 \times 90 = 900$.

Q15. Common ratio r = 6/2 = 3. Next terms: 54 × 3 = 162 and 162 × 3 = 486.
8th term = $2 \times 3^7 = 2 \times 2187 = 4374$.